Fundamentals of Human Resource Management (HRM1)

Module 4: HR Systems and Technology

Lesson 2: HR Technology and Artificial Intelligence

Lesson Notes

KMVTECHUG
Version: 2026

HRIS and Applicant Tracking Systems

A Human Resource Information System, commonly abbreviated HRIS, is a suite of software applications used to manage HR processes throughout the employee lifecycle. An HRIS enables an organisation to understand its workforce fully while staying compliant with changing tax laws and labour regulations. HRIS supports employee data management, payroll and benefits administration, time and attendance tracking, and compliance reporting, reducing manual labour and increasing data accuracy (Pay Wheel, 2024). By centralising employee information, HRIS also helps HR professionals make data driven decisions, revealing trends in performance, turnover, and recruitment outcomes that would otherwise be difficult to spot (International Journal of Multidisciplinary Research and Growth Evaluation, 2021). Examples of HRIS platforms include BambooHR, Microsoft Dynamics 365 for Talent, Oracle HCM Cloud, and Workday.

An Applicant Tracking System, commonly abbreviated ATS, automates recruitment specifically, handling job posting, resume screening, candidate ranking, interview scheduling, and recruitment reporting. Examples of ATS platforms include Oracle HCM Cloud Recruiting, iCIMS, Greenhouse, Taleo, and Workday Recruiting.

Where an organisation integrates its ATS and HRIS together, several benefits follow: improved data accuracy and compliance, streamlined HR operations, a better candidate and employee experience, automated onboarding and employee data creation, and seamless data transfer between recruitment and the rest of the HR system, rather than re-entering the same candidate information multiple times.

A Ugandan organisation still relying on spreadsheets for both recruitment tracking and employee records is, in effect, doing manually what an integrated ATS and HRIS would do automatically, at a significant cost in staff time and a higher risk of data entry errors.

> ***Reflect:*** *For a growing Ugandan organisation still using spreadsheets for HR, what would be the strongest argument for investing in a proper HRIS, even a relatively affordable one?*

AI Powered Chatbots in Recruitment, Onboarding and Employee Support

AI powered chatbots have transformed hiring by simplifying key steps, improving the applicant experience, and speeding up recruitment. Chatbots can review applications against predetermined standards such as required education, experience, and skills, removing the need for recruiters to manually sort through hundreds of submissions. Using natural language processing, chatbots match candidates to positions based on experience and skill evaluation, increasing recruiting speed and fit, while freeing recruiters to focus on higher value activities such as speaking directly with genuinely qualified candidates (Iqbal, 2018). Chatbots can also respond to candidates' questions about application status, job roles, and company policy, and some are capable of administering standardised personality assessments.

In onboarding, AI chatbots act as virtual assistants, helping new hires complete paperwork, fill out forms, and sign documents electronically, reducing administrative delays and freeing HR staff for more strategic work. Chatbots also support self paced learning by delivering training and orientation modules, monitoring progress, and providing immediate feedback, and can guide new employees through technical setup tasks such as device configuration and account creation (IBM, 2023; Schawbel, 2023).

In ongoing employee support, chatbots manage routine HR requests, such as questions about benefits or leave balances, making support available on demand at any hour. This accelerates response times and frees HR personnel for more complex work, while also promoting continual learning through training reminders tailored to an employee's role and goals (Schawbel, 2023; Johnson, 2022).

The overall advantages of AI chatbots in HR include cost effectiveness, scalability to handle many interactions simultaneously, and consistent, dependable information that reduces confusion among staff. A mid sized Ugandan organisation considering AI adoption in HR would likely see the clearest early benefit in handling routine employee questions automatically, freeing a small HR team to focus on more complex staff issues.

> ***Reflect:*** *Of recruitment, onboarding, and ongoing employee support, where do you think AI chatbots would add the most value for a Ugandan organisation with a very small HR team?*

Case Study, Google's Use of Workday, and HR Metrics and Analytics

Google uses Workday, a flexible, cloud based people management system, to improve and speed up HR tasks such as analytics, employee engagement, and talent allocation. Workday's human capital management tools facilitate employee lifecycle management from hiring through continuous development, while Workday Adaptive Planning allows Google to automate HR processes and analyse workforce data for strategic decisions. Tools such as Workday Prism Analysis and Workday Extend give Google access to real time data, strengthening its ability to manage workforce issues and talent effectively (ERP Today, 2021). Through this approach, Google aims to balance technological efficiency with a genuinely human centred HR strategy, reflecting a wider trend toward flexible, scalable digital HR solutions (Enterprise Times, 2023).

Beyond individual platforms like Workday, HR relies on metrics and analytics to guide decisions. HR metrics measure the efficiency of HR processes themselves, such as time to hire or employee turnover rate. HR analytics goes a step further, analysing that data to inform strategic decisions, such as predictive modelling of which employees are at risk of leaving, or talent pipeline analysis to anticipate future staffing gaps.

The integration of AI into HRIS platforms has added further efficiencies, including AI based training modules, automated candidate screening, and predictive analytics that evaluate staff turnover risk before it becomes a crisis, allowing HR teams to act proactively rather than reactively (Da Silva et al., 2022).

> ***Reflect:*** *If a Ugandan organisation could only adopt one HR technology capability right now, real time workforce data, AI chatbots, or predictive turnover analytics, which would you recommend, and why?*

Best Practices, Challenges and Future Trends in HR Technology

A few best practices apply broadly to HR system and technology adoption, regardless of an organisation's size. Align HR strategies clearly with organisational goals. Foster a genuinely positive work culture rather than treating culture as an afterthought. Invest consistently in employee development. Use technology deliberately to enhance HR processes rather than adopting it for its own sake. Monitor and evaluate HR metrics and analytics on an ongoing basis, not only during an annual review.

Organisations also face real challenges as HR technology evolves: managing remote and virtual teams effectively, closing skills gaps as talent needs shift, making genuinely data driven HR decisions rather than relying on instinct alone, advancing diversity, equity and inclusion initiatives meaningfully, and navigating digital transformation without simply adding technology on top of broken processes.

Looking ahead, several trends are shaping the future of HR technology: AI powered recruitment and HR processes becoming more common, mobile first and cloud based HR solutions becoming the norm rather than the exception, an increased focus on employee experience and engagement as a competitive differentiator, deeper integration with emerging technologies, and enhanced analytics with genuinely predictive capabilities.

Closing thought for this module: HR systems and technology exist to serve the people and processes described in Lesson 1, not to replace sound HR judgement. The best organisations use technology to remove administrative friction, freeing HR professionals to focus on the strategic and human aspects of the role that no software can fully replicate.